

Task 6: Sales Trend Analysis Using Aggregations

Project README & Execution Documentation • MySQL Workbench

INTERN / AUTHOR Rushikesh Rathod	TOOL / ENVIRONMENT MySQL Workbench 8.0	PRIMARY DATASET salesdataset.csv (15,030 rows)	STATUS Completed & Verified
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15,030 TOTAL RECORDS ANALYZED	\$44,506,652.27 CUMULATIVE GROSS REVENUE	36 Months TIME SPAN (2023 - 2025)
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1. Project Objective

The primary goal of Task 6 is to evaluate and identify seasonal patterns, customer purchasing cycles, and business growth trends by analyzing monthly revenue and transaction order volume using SQL aggregation functions.

Core Deliverables:

- Production-ready SQL script file (task_6_sales_trend_analysis_Rushikesh_Rathod.sql).
- Tabular results table demonstrating aggregate order volume and sum of revenue per month.

2. Step-by-Step Implementation Workflow

- Database & Schema Setup:** Established the schema sales_project in MySQL Workbench and utilized the *Table Data Import Wizard* to ingest salesdataset.csv into the table online_sales.
- Date Handling & Sanitization:** Handled string-formatted dates (DD-Mon-YY) dynamically using MySQL's native STR_TO_DATE(OrderDate, '%d-%b-%y') function to ensure robust, fault-tolerant date parsing.
- Temporal Feature Extraction:** Utilized EXTRACT(YEAR FROM ...) and EXTRACT(MONTH FROM ...) to isolate calendar dimensions.
- Metric Aggregations:** Aggregated unique transaction counts using COUNT(DISTINCT OrderID) and summed commercial financial value using ROUND(SUM(TotalAmount), 2).
- Grouping & Chronological Sorting:** Applied GROUP BY sales_year, sales_month and sorted sequentially using ORDER BY sales_year ASC, sales_month ASC.

3. SQL Implementation Scripts

Query 1: Complete Historical Monthly Aggregations (2023 - 2025)

```
-- Historical Monthly Revenue and Volume Analysis
SELECT
    EXTRACT(YEAR FROM STR_TO_DATE(OrderDate, '%d-%b-%y')) AS sales_year,
    EXTRACT(MONTH FROM STR_TO_DATE(OrderDate, '%d-%b-%y')) AS sales_month,
    COUNT(DISTINCT OrderID) AS total_orders,
    ROUND(SUM(TotalAmount), 2) AS total_revenue
FROM
    online_sales
WHERE
    OrderDate IS NOT NULL
GROUP BY
    sales_year,
    sales_month
ORDER BY
    sales_year ASC,
    sales_month ASC;
```

Query 2: Time-Period Specific Analysis (Year 2024 per Hint 'f')

```
-- Filtered Analysis for Fiscal Year 2024
SELECT
    EXTRACT(YEAR FROM STR_TO_DATE(OrderDate, '%d-%b-%y')) AS sales_year,
    EXTRACT(MONTH FROM STR_TO_DATE(OrderDate, '%d-%b-%y')) AS sales_month,
    COUNT(DISTINCT OrderID) AS total_orders,
    ROUND(SUM(TotalAmount), 2) AS total_revenue
FROM
    online_sales
WHERE
    STR_TO_DATE(OrderDate, '%d-%b-%y') BETWEEN '2024-01-01' AND '2024-12-31'
GROUP BY
    sales_year,
    sales_month
ORDER BY
    sales_month ASC;
```

4. Results Table Preview (Year 2024 Aggregations)

Sales Year	Month	Total Distinct Orders	Total Revenue (USD)	Status / Trend Note
2024	1 (Jan)	441	\$1,211,584.10	Q1 Steady Base
2024	2 (Feb)	383	\$1,165,342.09	Low Volume Period
2024	3 (Mar)	414	\$1,225,723.59	Q1 Recovery
2024	4 (Apr)	413	\$1,186,125.04	Moderate Stability
2024	5 (May)	430	\$1,248,744.46	Mid-Year Growth
2024	6 (Jun)	400	\$1,116,894.83	Trough Volume

Sales Year	Month	Total Distinct Orders	Total Revenue (USD)	Status / Trend Note
2024	7 (Jul)	440	\$1,289,434.25	Q3 Upward Trend
2024	8 (Aug)	465	\$1,451,380.71	Peak Annual Revenue
2024	9 (Sep)	437	\$1,419,788.39	High Demand Month
2024	10 (Oct)	428	\$1,229,952.42	Normalizing Run Rate
2024	11 (Nov)	368	\$1,136,701.74	Pre-Holiday Drip
2024	12 (Dec)	435	\$1,339,673.17	Holiday Surge

5. Key Analytical Insights

- **Peak Performance:** August 2024 emerged as the highest revenue-generating month (\$1,451,380.71 across 465 transactions), followed closely by September 2024.
- **Annual Consistency:** The organization averaged approximately \$1.25M in revenue and ~420 orders per month, indicating resilient customer retention across global sales channels.
- **Seasonality:** Significant surges are observed in late Q3 (August-September) and late Q4 (December holiday sales).